

RenderSurfaceLighting()

```
for all lights do  
    Clear Depth Texture  
    for all surfaces do  
        Generate MVP  
        WriteDepthBuffer()  
    end for  
end for
```

WriteDepthBuffer()

```
Set shader program patch_tess_pass  
Assign SSBO's and uniforms, light_pass = 1  
for all surface control points (concurrently) do  
    Dispatch compute shader patch_tess_pass  
    Get a triangulated version of our patch control points as an EBO  
end for  
Set shader program patch_shadow  
Render to depth buffer the triangulation of the control points  
  
if visualize then  
    Set shader program visualize_tiles  
    Render triangulated tiles to primary camera  
end if
```

patch_tess_pass: light pass

```
Interpret BB-coefficient/control point vertex data  
Calculate 2nd derivative in the U direction  
Calculate 2nd derivative in the V direction  
for all tiles do  
    Get indices of its two triangles  
    Place those six indices into the EBO with control points as vertices  
    (triangulating shadow mapping)  
end for
```

```

patch_shadow
for all vertices do
    Project from light source
    Save projection of control point triangulation to light's depth buffer
end for

```

```

RenderSurfaces()
for all surfaces do
    UpdatePatchTessLevels()
    RenderSurface()
end for

```

```

UpdatePatchTessLevels()
Set shader program patch_tess_pass
Assign SSBO's and uniforms, light_pass = 0
for all surface control points (concurrently) do
    Dispatch compute shader patch_tess_pass
end for

```

```

patch_tess_pass: tess pass
Interpret BB-coefficient/control point vertex data
Calculate 2nd derivative in the U direction
Calculate 2nd derivative in the V direction
for all patches do
    for all 9 bi-1 quads of a bi-3 BB-net, simultaneously do
        Compute center box
    end for
    Create average center box of all four control point min-max boxes
    Let A = All five rectangles that enclose projected boxes
    Get Max(max_A x_length, max_A y_length)
    if patch is entirely out of view then
        Set tessellation factor to 0.0
    end if
end for

```

RenderSurface()

Set shader program *render_pass*

Assign SSBO's and uniforms

Draw Elements (render_pass.vs, .tcs, .tes, .fs)

\\IMPORTANT

\\Tessellation factors are assigned in Tess Control shader

\\Fragment shader reads depth buffer to create shadows